COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 007	**SECTION:**	CPE11S4
DATE PERFORMED:	9/11/2026	**DATE SUBMITTED:**	9/11/2026
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ACTIVITY NO. 4.2 Characters and Strings

1. PROCEDURE OUTPUT

1. Try to create a program that outputs the following:

```
Testing islower:
p is a lowercase letter
P is not a lowercase letter
5 is not a lowercase letter
! is not a lowercase letter
Testing isupper:
D is an uppercase letter
d is not an uppercase letter
8 is not an uppercase letter
& is not an uppercase letter
u converted to uppercase is U
7 converted to uppercase is 7
$ converted to uppercase is $
L converted to lowercase is l
```

Activity > 4.2.cpp > main()

```cpp
#include <iostream>
#include <cstring>
#include <string>
#include <cctype>
using namespace std;

int main() {
// Testing is lower

char ch; //This is a character Constant
cout << "Enter four(4) character: \n";
cout << "Testing islower()\n";
for (int i=0;i<4;i++){
    cout << "Enter a character: ";
    cin >> ch;

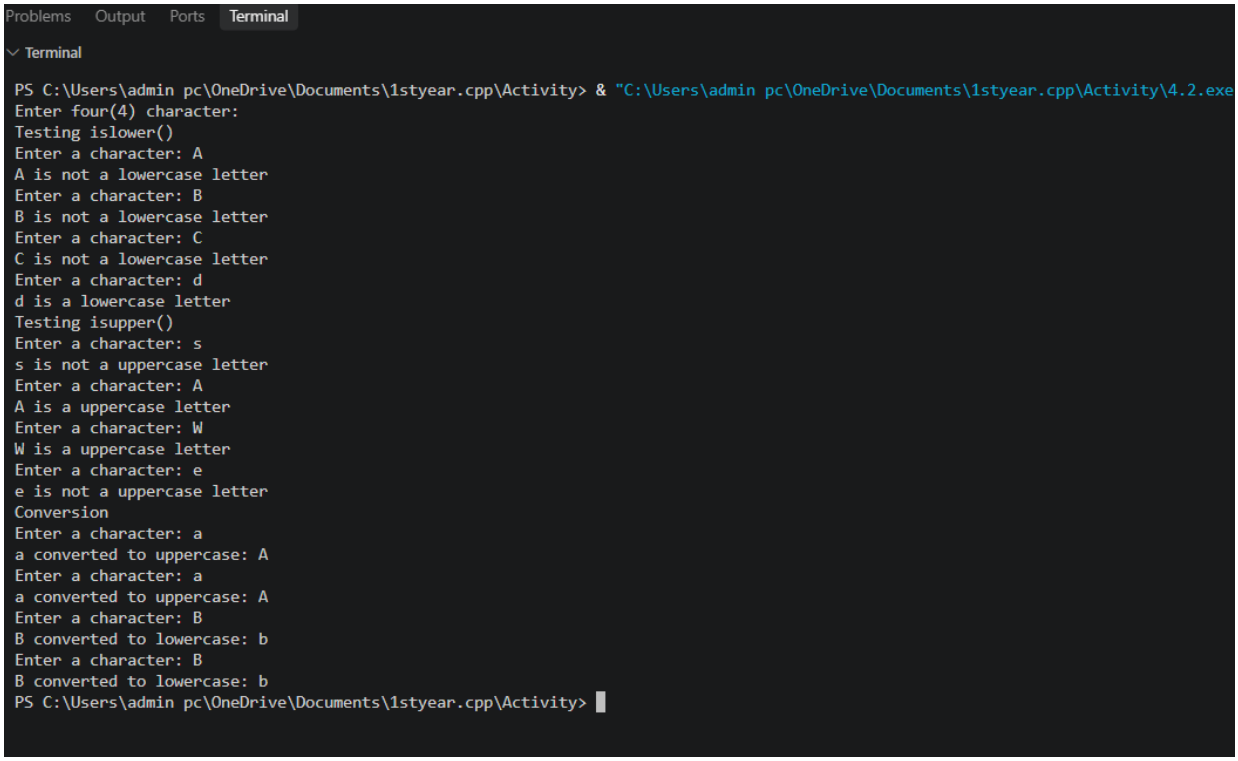
    if(islower(ch)){
        cout << ch << " is a lowercase letter\n";
    } else {
        cout << ch << " is not a lowercase letter\n";
    }
}

cout << "Testing isupper()\n";
for (int i=0;i<4;i++){
    cout << "Enter a character: ";
    cin >> ch;

    if(isupper(ch)){
        cout << ch << " is a uppercase letter\n";
    } else {
        cout << ch << " is not a uppercase letter\n";
    }
}

cout << "Conversion\n";
for (int i=0;i<4;i++){
    cout << "Enter a character: ";
    cin >> ch;

    if(islower(ch)){
        cout << ch << " converted to uppercase: " << (char)toupper(ch) << endl;
    }else if (isupper(ch)){
        cout << ch << " converted to lowercase: " << (char)tolower(ch) << endl;
    } else {
        cout << ch << " is not convertible. " << endl;
    }
}
return 0;
}
```



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

For each of the following, Create the following C++ Code and output for each case:

1. Write a C++ program that accepts a character input from the keyboard and tests the character using each function in the <cctype> character-handling library. Display the result of each function.

Activity > 4.2.cpp > main()

```cpp
#include <iostream>
#include <cctype>
using namespace std;

int main() {
    char ch;

    cout << "Enter a character: ";
    cin >> ch;

    cout << endl;

    if (isalpha(ch))
        cout << "The character is an alphabet." << endl;
    else
        cout << "The character is not an alphabet." << endl;

    if (isdigit(ch))
        cout << "The character is a digit." << endl;
    else
        cout << "The character is not a digit." << endl;

    if (isalnum(ch))
        cout << "The character is alphanumeric." << endl;
    else
        cout << "The character is not alphanumeric." << endl;

    if (islower(ch))
        cout << "The character is lowercase." << endl;
    else
        cout << "The character is not lowercase." << endl;

    if (isupper(ch))
        cout << "The character is uppercase." << endl;
    else
        cout << "The character is not uppercase." << endl;

    if (isspace(ch))
        cout << "The character is a whitespace." << endl;
    else
        cout << "The character is not a whitespace." << endl;

    if (isblank(ch))
        cout << "The character is a blank." << endl;
```

```
    else
        cout << "The character is not a blank." << endl;

    if (isprint(ch))
        cout << "The character is printable." << endl;
    else
        cout << "The character is not printable." << endl;

    if (isgraph(ch))
        cout << "The character is graphical." << endl;
    else
        cout << "The character is not graphical." << endl;

    if (ispunct(ch))
        cout << "The character is a punctuation character." << endl;
    else
        cout << "The character is not a punctuation character." << endl;

    cout << "The lowercase version is: " << char(tolower(ch)) << endl;
    cout << "The uppercase version is: " << char(toupper(ch)) << endl;

    return 0;
}
```

OUTPUT

```
Problems  Output  Ports  Terminal

Terminal

PS C:\Users\admin pc\OneDrive\Documents\1styear.cpp\Activity> & "C:\Users\admin pc\One
ar.cpp\Activity\4.2.exe"
Enter a character: M

The character is an alphabet.
The character is not a digit.
The character is alphanumeric.
The character is not lowercase.
The character is uppercase.
The character is not a whitespace.
The character is not a blank.
The character is printable.
The character is graphical.
The character is not a punctuation character.
The lowercase version is: m
The uppercase version is: M
PS C:\Users\admin pc\OneDrive\Documents\1styear.cpp\Activity>
```

2. Write a C++ program that accepts four strings representing integers, converts them into integer values, calculates their sum, and displays the total.

Activity > 4.2.cpp > main()

```cpp
#include <iostream>
#include <string>
using namespace std;

int main() {
    string num1, num2, num3, num4;
    int sum;

    cout << "Enter first integer: ";
    cin >> num1;

    cout << "Enter second integer: ";
    cin >> num2;

    cout << "Enter third integer: ";
    cin >> num3;

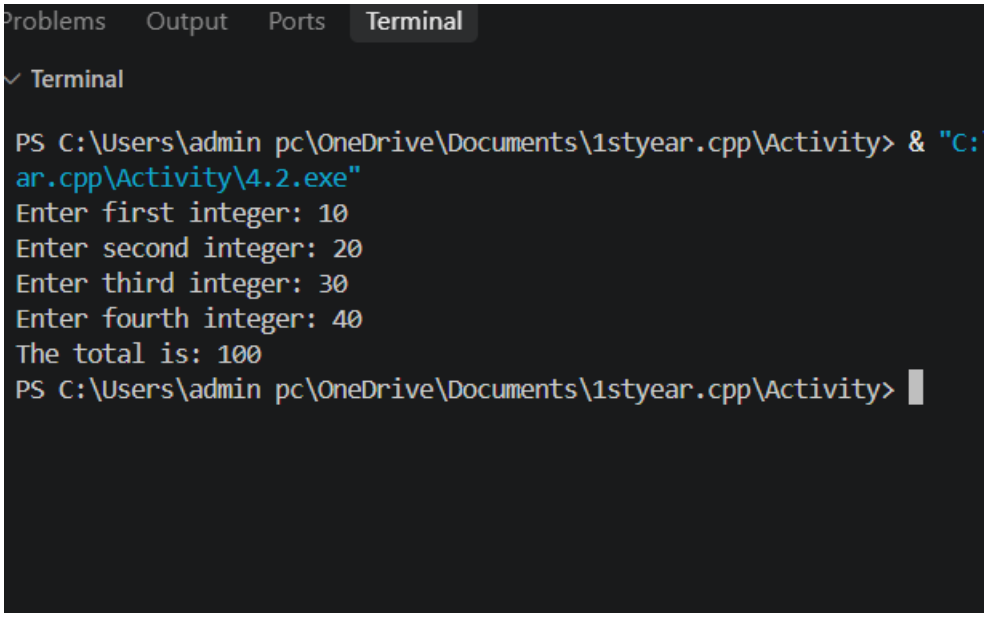
    cout << "Enter fourth integer: ";
    cin >> num4;
//stoi ay para maging integer yung string na input
    int n1 = stoi(num1);
    int n2 = stoi(num2);
    int n3 = stoi(num3);
    int n4 = stoi(num4);

    sum = n1 + n2 + n3 + n4;

    cout << "The total is: " << sum << endl;

    return 0;
}
```

OUTPUT



3. Write a C++ program that accepts a password as a string and determines whether it meets all of the following requirements:

- Contains at least one uppercase letter
- Contains at least one lowercase letter
- Contains at least one digit
- Contains at least one punctuation character
- Contains no whitespace characters
- Contains at least 8 characters

The program must use appropriate functions from the <cctype> library to examine each character. Display which requirements are satisfied or not satisfied, and indicate whether the password is VALID or INVALID.

Activity > 4.2.cpp > main()

```cpp
#include <iostream>
#include <string>
#include <cctype>
using namespace std;

int main() {
    string password;

    int upper = 0;
    int lower = 0;
    int digit = 0;
    int punct = 0;
    int space = 0;

    cout << "Enter password: ";
    cin >> password;

    for (int i = 0; i < password.length(); i++) {

        if (isupper(password[i]))
            upper = 1;

        if (islower(password[i]))
            lower = 1;

        if (isdigit(password[i]))
            digit = 1;

        if (ispunct(password[i]))
            punct = 1;

        if (isspace(password[i]))
            space = 1;
    }

    cout << endl;

    if (upper == 1)
        cout << "Uppercase letter: SATISFIED" << endl;
    else
        cout << "Uppercase letter: NOT SATISFIED" << endl;

    if (lower == 1)
        cout << "Lowercase letter: SATISFIED" << endl;
```

```
45      else
46          cout << "Lowercase letter: NOT SATISFIED" << endl;
47
48      if (digit == 1)
49          cout << "Digit: SATISFIED" << endl;
50      else
51          cout << "Digit: NOT SATISFIED" << endl;
52
53      if (punct == 1)
54          cout << "Punctuation: SATISFIED" << endl;
55      else
56          cout << "Punctuation: NOT SATISFIED" << endl;
57
58      if (space == 0)
59          cout << "No whitespace: SATISFIED" << endl;
60      else
61          cout << "No whitespace: NOT SATISFIED" << endl;
62
63      if (password.length() >= 8)
64          cout << "At least 8 characters: SATISFIED" << endl;
65      else
66          cout << "At least 8 characters: NOT SATISFIED" << endl;
67
68      if (upper == 1 && lower == 1 && digit == 1 &&
69          punct == 1 && space == 0 && password.length() >= 8)
70          cout << "\nPassword is VALID." << endl;
71      else
72          cout << "\nPassword is INVALID." << endl;
73
74      return 0;
75  }
```

OUTPUT

```
PS C:\Users\admin pc\OneDrive\Documents\1styear.cpp\Activity> & "C:\Users\admin pc\OneDrive\Documents\1styear.cpp\Activity\4.2.exe"
Enter password: Hello123!

Uppercase letter: SATISFIED
Lowercase letter: SATISFIED
Digit: SATISFIED
Punctuation: SATISFIED
No whitespace: SATISFIED
At least 8 characters: SATISFIED

Password is VALID.
PS C:\Users\admin pc\OneDrive\Documents\1styear.cpp\Activity>
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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